Seventh Section.

Reality of the Roots.

§ 82. Quadratic forms with non-vanishing determinant

In the investigation of the forms $\varphi(x_1, x_2, \ldots, x_m)$ with non-vanishing determinant, we make use of a theorem on determinants that we proved at the end of § 28, and which we recall here.

If A is a determinant of m rows, and if $A_i^{(k)}$ are its first and $A_{i,k}^{h,l}$ its second minors, then

$$A_1^{(1)} A_i^{(i)} - A_1^{(i)} A_i^{(1)} = A A_{1,i}^{1,i}. \tag{1}$$

If A is a non-zero symmetric determinant, then $A_1^{(i)} = A_i^{(1)}$, and we may write (1) in the form

$$A_1^{(1)} A_i^{(i)} - \left(A_1^{(i)}\right)^2 = A A_{1,i}^{1,i}. \tag{2}$$

Here i may denote any one of the indices $2, 3, \ldots, m$. If $A_1^{(1)} = 0$ and $A_{1,i}^{1,i} = 0$, it follows that also $A_1^{(i)} = 0$; from this we conclude that the quantities

$$A_1^{(1)}, \quad A_{1,2}^{1,2}, \quad A_{1,3}^{1,3}, \ldots, A_{1,m}^{1,m}$$

cannot all vanish at the same time, for then

$$A_1^{(2)}, \quad A_1^{(3)}, \ldots, A_1^{(m)},$$

and hence, contrary to the assumption, A itself would vanish (§ 22).

We now apply this to the determinant $R = R_m$ of our function φ, which is assumed non-zero. It follows that although the first principal minor R_{m-1} may vanish, not all second principal minors R_{m-2} can vanish. The same argument can be applied when we replace R by a non-zero $R_{m-1}, R_{m-2}, \ldots$. We therefore obtain the following important theorem:

III. If R is non-zero, the indices $1, 2, \ldots, m$ can be ordered so that in the chain of principal minors

$$R_m, R_{m-1}, \ldots, R_1, R_0 \tag{3}$$

no two consecutive terms vanish.

The determinant relation (2), when one puts

$$A = R_{k+1}, \qquad (k = 1, 2, \ldots, m-1),$$

gives an equation between three successive terms of the chain (3), which we may write as

$$R_k S_k - T_k^2 = R_{k-1} R_{k+1}, \tag{4}$$

where S_k and T_k are certain minors of R, and hence integral rational functions of the coefficients $a_{i,k}$.

More precisely, R_k, S_k, T_k are first minors of R_{k+1}, namely

$$R_k = \frac{\partial R_{k+1}}{\partial a_{k+1,k+1}}, \qquad S_k = \frac{\partial R_{k+1}}{\partial a_{k,k}}, \qquad T_k = \frac{\partial R_{k+1}}{\partial a_{k,k+1}}.$$

We shall suppose in what follows that an ordering of the indices corresponding to theorem (3) has been chosen. Then, if R_k vanishes, R_{k-1} and R_{k+1} are non-zero, and (4) shows that they have opposite signs. Thus:

IV. If an interior term of a chain of principal minors vanishes, then the two adjacent terms have opposite signs.

§ 83. Number of positive and negative squares

To determine the number of positive and negative squares of a form with non-vanishing determinant, we first suppose that no term vanishes in the chain of principal minors

$$R_m, R_{m-1}, R_{m-2}, \ldots, R_1, R_0 = 1.$$

We can then choose the coefficient λ so that the determinant of the form

$$\psi = \varphi(x_1, x_2, x_3, \ldots, x_m) - \lambda x_m^2 \tag{1}$$

vanishes. Its determinant is in fact

$$\begin{vmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,m} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,m} \\ \cdots & \cdots & & \cdots \\ a_{m,1} & a_{m,2} & \cdots & a_{m,m} - \lambda \end{vmatrix} = R_m - \lambda R_{m-1},$$

and it vanishes when

$$\lambda = \frac{R_m}{R_{m-1}}$$

is put.

The function ψ can then be expressed by means of $m-1$ variables y, and by formula (6), § 81, one obtains

$$\psi(x_1, x_2, \ldots, x_m) = \psi(y_1, y_2, \ldots, y_{m-1}, 0),$$

or, by (1),

$$\varphi(x_1, x_2, \ldots, x_m) = \frac{R_m}{R_{m-1}} x_m^2 + \varphi(y_1, y_2, \ldots, y_{m-1}, 0). \tag{2}$$

Thus the investigation of the function φ in m variables is reduced to that of

$$\varphi_1(y) = \varphi(y_1, y_2, \ldots, y_{m-1}, 0)$$

in $m-1$ variables, whose determinant is equal to R_{m-1} and is therefore non-zero. According as $R_m : R_{m-1}$ is positive or negative, the function $\varphi(x)$ has one positive or one negative square more than $\varphi_1(y)$.

Applying the same procedure to $\varphi_1(y)$ and to the following functions gives the theorem:

The number of positive and negative terms of the sequence

$$\frac{R_m}{R_{m-1}}, \quad \frac{R_{m-1}}{R_{m-2}}, \quad \ldots, \quad \frac{R_1}{R_0} \tag{3}$$

agrees with the number of positive and negative squares into which the function $\varphi(x)$ can be decomposed.

We shall now give this theorem a somewhat different form, but first introduce the following terminology. If a sequence of non-zero real numbers is given in a definite order, their signs may change in various ways. If two quantities of the same sign follow one another, there is a permanence of sign; but if a quantity is followed by another with the opposite sign, there is a variation of sign.

Now consider from this point of view the sequence of quantities

$$R_m, R_{m-1}, R_{m-2}, \ldots, R_1, R_0.$$

In passing from R_k to R_{k-1} one obtains a permanence or a variation of sign according as the quotient $R_k : R_{k-1}$ is positive or negative.

We may therefore also state the following theorem:

V. If π is the number of positive squares and ν the number of negative squares of φ, then π is equal to the number of permanences, and ν to the number of variations, in the chain

$$R_m, R_{m-1}, R_{m-2}, \ldots, R_1, R_0. \tag{4}$$

This form of the theorem has the advantage that it carries over to the case where individual interior terms in (4) vanish, provided that no two consecutive terms are zero, which by § 82, III can always be assumed. If, for instance, $R_k = 0$, while R_{k+1} and R_{k-1} are non-zero, then by § 82, theorem IV, R_{k-1} and R_{k+1} have opposite signs. In the sequence

$$R_{k+1}, \quad R_k, \quad R_{k-1}$$

there is therefore one variation and one permanence of sign, no matter whether we replace the vanishing R_k by a positive or by a negative quantity.

If we replace the chain (4)

$$R_m, R_{m-1}, \ldots, R_1, R_0,$$

in which individual terms vanish, by another chain

$$R'_m, R'_{m-1}, \ldots, R'_1, R'_0, \tag{5}$$

in which no term vanishes, and in which the non-vanishing terms of (4) are matched by terms of the same sign, then (4) and (5) have the same number of variations, whatever signs the R' corresponding to the vanishing R may have.

Let $\psi(x)$ now be a second arbitrary quadratic form in the variables x, and form

$$\varphi' = \varphi + \varepsilon\psi, \tag{6}$$

where ε is a coefficient still to be determined. We shall show at once that ε can be so chosen that φ' and φ have the same number of positive and negative squares, that the chain of principal minors R'_k of φ' contains no vanishing term, and finally that a non-vanishing R_k corresponds to an R'_k of the same sign. Then the numbers π, ν for φ and φ' can be determined from (4) as well as from (5), and the number of variations, which is the same in both, gives the number ν of negative squares.

To prove that ε can be chosen in this way, suppose that φ has somehow been transformed into a sum of squares,

$$\varphi = \lambda_1 y_1^2 + \lambda_2 y_2^2 + \cdots + \lambda_m y_m^2,$$

and that ψ, expressed in the variables $y_1, y_2, \ldots, y_m$, has the form

$$\psi = \sum \beta_{i,k} y_i y_k.$$

The numbers π, ν for the function φ' are then determined, by theorem V, from the chain of principal minors of

$$\begin{vmatrix} \lambda_1 + \varepsilon\beta_{1,1} & \varepsilon\beta_{1,2} & \cdots & \varepsilon\beta_{1,m} \\ \varepsilon\beta_{2,1} & \lambda_2 + \varepsilon\beta_{2,2} & \cdots & \varepsilon\beta_{2,m} \\ \cdots & \cdots & & \cdots \\ \varepsilon\beta_{m,1} & \varepsilon\beta_{m,2} & \cdots & \lambda_m + \varepsilon\beta_{m,m} \end{vmatrix}.$$

But ε can be taken so small that these principal minors agree in sign with

$$\lambda_1\lambda_2\lambda_3\cdots\lambda_m, \quad \lambda_1\lambda_2\lambda_3\cdots\lambda_{m-1}, \quad \ldots, \quad \lambda_1\lambda_2, \quad \lambda_1, \quad 1.$$

Thus the number of positive and negative squares of φ' is equal to the number of positive and negative coefficients λ; none of these vanishes. Hence the numbers π and ν are the same for φ and φ'.

Now let the coefficients of ψ, expressed in the original variables, be $b_{i,k}$, so that

$$\psi = \sum b_{i,k} x_i x_k.$$

Then one of the principal minors of φ' is

$$R'_k = \begin{vmatrix} a_{1,1} + \varepsilon b_{1,1} & \cdots & a_{1,k} + \varepsilon b_{1,k} \\ \cdots & & \cdots \\ a_{k,1} + \varepsilon b_{k,1} & \cdots & a_{k,k} + \varepsilon b_{k,k} \end{vmatrix},$$

and is therefore an integral rational function of degree k in ε,

$$R'_k = R_k + \varepsilon M_1 + \varepsilon^2 M_2 + \cdots + \varepsilon^k M_k,$$

where $M_1, M_2, \ldots, M_k$ depend rationally on the $a_{i,k}, b_{i,k}$. In particular,

$$M_k = \sum \pm b_{1,1} b_{2,2} \cdots b_{k,k},$$

and the $b_{i,k}$ can always be chosen so that M_k is non-zero. By § 31 one can therefore choose ε so that, if R_k is non-zero, R'_k has the same sign as R_k, and if R_k vanishes, R'_k does not vanish.

Combining what has just been proved with the results of § 81, we obtain the following general rule for determining the number of negative squares, also in the case of a vanishing determinant.

VI. To determine the numbers π, ν, ρ of the function $\varphi(x)$, arrange the variables $x_1, x_2, \ldots, x_m$ so that in the chain of principal minors

$$R_m, R_{m-1}, \ldots, R_1, R_0 \tag{7}$$

as small as possible a number of initial terms vanish, and so that among the following terms no two adjacent terms vanish. Then ρ is the number of vanishing initial terms, ν the number of variations of sign, and

$$\pi = m - \nu - \rho.$$

Finally, it should be remarked that the number of variations in the sequence (7) can also be counted from right to left; that is, one finds the same number of variations when the sequence is written in the reverse order.

§ 84. Application to the Bezoutian

The detailed discussion of the inertia of quadratic forms has had only one purpose for us: to determine the number of real and imaginary roots of an algebraic equation by counting the positive and negative squares of the Bezoutian. If n is the degree of the equation, its Bezoutian is a quadratic form in $n - 1$ variables, and in § 73 we constructed it completely for $n = 3, 4, 5$ and learned the methods by which one may also compute it in other cases.

We have already noted that the Bezoutian is not a general quadratic form. It has the special property that the number ν of its negative squares can never be larger than $\frac{1}{2}n$. This property must find its expression in certain inequalities between the coefficients, but these are not yet known. Only in the case $n = 3$ can we give the algebraic character of this restriction completely.

For the cubic equation

$$f(x) = a_0 x^3 + a_1 x^2 + a_2 x + a_3 = 0$$

the Bezoutian, by § 73, (13), is nothing other than the Hessian covariant

$$-\frac{2}{3} H(t_1, t_0) = -\frac{2}{3} \left(A_0 t_1^2 + A_1 t_1 t_0 + A_2 t_0^2 \right).$$

But if D denotes the discriminant, so that

$$3D = 4A_0 A_2 - A_1^2,$$

then the relation [§ 62, (4), (12)]

$$4H^3 + Q^2 + 27Df^2 = 0$$

holds. Therefore, if we put $t_0 = 0$, we obtain

$$-4A_0^3 + q_0^2 + 27Da_0^2 = 0, \tag{1}$$

where

$$q_0 = 27a_0^2a_3 - 9a_0a_1a_2 + 2a_1^3.$$

This relation shows that A_0 and D cannot both be positive, and therefore that the form $-H$ cannot be decomposed into two negative squares.

We shall now apply the general theorem of the preceding paragraph to the biquadratic equation, which for simplicity we take in the form

$$x^4 + ax^2 + bx + c = 0. \tag{2}$$

To form the coefficients of the Bezoutian, we must replace, in the formulas § 73, (14),

$$a_0, a_1, a_2, a_3, a_4$$

by

$$1, 0, a, b, c.$$

Thus we obtain

$$\begin{array}{lll} B_{2,2} = -2a, & B_{1,1} = a^2 - 4c, & B_{0,0} = -\frac{1}{4}(8ac - 3b^2), \\ B_{0,1} = \frac{1}{2}ab, & B_{2,0} = -4c, & B_{1,2} = -3b. \end{array}$$

We order the determinant as follows:

$$\begin{vmatrix} -2a & -3b & -4c \\ -3b & a^2 - 4c & \frac{1}{2}ab \\ -4c & \frac{1}{2}ab & -\frac{1}{4}(8ac - 3b^2) \end{vmatrix},$$

and obtain the chain of principal minors

$$D, \quad -2a^3 + 8ac - 9b^2, \quad -2a, \quad 1, \tag{3}$$

where D denotes the discriminant, which by § 64, (11) is determined by

$$27D = 4(a^2 + 12c)^3 - (2a^3 - 72ac + 27b^2)^2. \tag{4}$$

The criterion that all four roots be real is therefore

$$D > 0, \qquad -2a^3 + 8ac - 9b^2 > 0, \qquad -2a > 0. \tag{5}$$

This criterion seems different, and in any case less simple, than the one established in § 78,

$$D > 0, \qquad a < 0, \qquad a^2 - 4c > 0. \tag{6}$$

We shall now show, however, that the two say exactly the same thing.

It is first immediately clear that (6) must hold when (5) holds, for

$$-2a(a^2 - 4c) - 9b^2$$

cannot be positive if a and $a^2 - 4c$ are negative. Conversely, to see that (6) implies (5), we must take the value of D into consideration.

For brevity put

$$\alpha = -2a, \qquad \beta = -6a^3 + 24ac - 27b^2, \qquad \gamma = 3a^2 - 12c. \tag{7}$$

Then the conditions (5) are

$$\alpha > 0, \qquad \beta > 0, \qquad D > 0,$$

while the conditions (6) are

$$\alpha > 0, \qquad \gamma > 0, \qquad D > 0.$$

It remains to prove that if α, γ, and D are positive, then β must also be positive. To this end we express D in (4), by means of (7), in terms of α, β, γ and obtain

$$27D = 4(\alpha^2 - \gamma)^3 - \left[2\alpha(\alpha^2 - \gamma) - \beta\right]^2.$$

If D is positive, then necessarily $\alpha^2 - \gamma$ is positive. Put

$$\alpha^2 - \gamma = \delta^2.$$

Then $\alpha^2 > \delta^2$, hence, if δ is taken positive,

$$\alpha > \delta,$$

and

$$27D = 4\delta^6 - (2\alpha\delta^2 - \beta)^2.$$

Thus D cannot be positive when β is negative, for if β is negative, then

$$2\alpha\delta^2 - \beta > 2\alpha\delta^2 > 2\delta^3,$$

and therefore $4\delta^6 - (2\alpha\delta^2 - \beta)^2$ is negative.

This directly proves that the criteria (5) and (6) have exactly the same meaning.

According to the geometric interpretation given in § 78, $\beta = 0$ denotes a surface of third order that passes through the parabola $b = 0, \gamma = 0$, and, insofar as negative values of a are concerned, lies entirely in the part of space in which D is negative.

To have before us an example of a quintic equation, consider

$$x^5 + x^2 + a = 0.$$

Thus in our general expressions § 73, or also § 74, (2), (3), we have to put

$$a_0, a_1, a_2, a_3, a_4, a_5 = 1, 0, 0, 1, 0, a,$$

and for the determinant of the Bezoutian we obtain

$$\begin{vmatrix} -2a & 0 & 0 & -5a \\ 0 & \frac{6}{5} & -5a & 0 \\ 0 & -5a & 0 & -3 \\ -5a & 0 & -3 & 0 \end{vmatrix},$$

and for the discriminant

$$D = 108a + 3125a^4.$$

A chain of principal minors is

$$D, \quad 50a^3, \quad \frac{-12a}{5}, \quad -2a, \quad 1. \tag{8}$$

The discriminant is negative when

$$0 < -a < \sqrt[3]{\frac{108}{3125}}, \tag{9}$$

and positive otherwise, in particular for all positive a. One sees that when the discriminant is positive, (8) always has two variations of sign, while for a negative discriminant, where a is also negative, it has one variation.

Our equation therefore has two imaginary roots as long as a lies in the interval (9), and four imaginary roots when a lies outside this interval; it never has four real roots.

A purely numerical example is provided by the equation, arising from the complex multiplication of elliptic functions,

$$x^5 - x^3 - 2x^2 - 2x - 1 = 0.$$

For the determinant of the Bezoutian we obtain from § 73

$$\begin{vmatrix} -\frac{4}{5} & -\frac{3}{5} & \frac{4}{5} & 5 \\ -\frac{3}{5} & \frac{4}{5} & \frac{33}{5} & 8 \\ \frac{4}{5} & \frac{33}{5} & \frac{46}{5} & 6 \\ 5 & 8 & 6 & 2 \end{vmatrix},$$

and the chain of principal minors

$$\frac{47^2}{5}, \quad \frac{94}{5}, \quad -1, \quad -\frac{4}{5}, \quad 1.$$

Thus the discriminant of our equation is 47^2. The equation therefore has two pairs of imaginary roots and one real root. If the discriminant is known, the second term of this chain, $94 : 5$, need not be computed; indeed, it is already enough to know the negative sign of the penultimate term in order to decide the reality of the roots as above.

Eighth Section.

Sturm's Theorem.

§ 85. Sturm's problem

The question treated in the preceding section, namely the number of real roots of an equation, is a special case of a more general problem. This problem will form the subject of the present section, and it is the foundation of all methods for the approximate numerical calculation of roots of equations.

The question is this: how many real roots of a real numerical equation lie between two given real numerical values a and b?

Once this has been decided, the next task is to narrow the bounds a, b until only one root of the given equation lies between them, and finally to bring them so close together that either of them may be regarded as an approximate value of that root.

If we take $a = -\infty, b = +\infty$, or at least a negative and b positive so large that no roots can lie beyond these bounds, then this task coincides with the one treated in the preceding section: the determination of the number of all real roots.

We first show how the general problem can be reduced to the special one, so that the question just posed is answered, at least in principle, by the considerations of the preceding section. Suppose first that we wish to determine the number of positive roots of an equation $f(x) = 0$. Put $x = y^2$. Then to every real value of y there corresponds a positive value of x, and conversely to every positive value of x there correspond two real values of y with opposite signs. The number of positive roots of $f(x) = 0$ is therefore half the number of real roots of $f(y^2) = 0$.

Further, put

$$y = \frac{x-a}{b-x}.$$

Then, as x goes from a to b, y runs through positive values from 0 to infinity; and if under this substitution $f(x)$ is transformed into $F(y)$, the equation $f(x) = 0$ has as many roots between a and b as $F(y) = 0$ has positive roots, hence half as many as $F(z^2) = 0$ has real roots. Thus our problem is indeed reduced to determining the number of real roots of a certain other equation, whose degree is twice as large as the degree of the given equation. But this reduction does not give the solution in the simplest form, and we must seek a simpler answer to the question.

§ 86. Sturm chains

The following consideration leads to a solution of the problem set in the preceding paragraph.

Let α and β be any two real numbers with $\alpha < \beta$. Let $f(x)$ be a given integral rational function of x, and suppose that we are to determine how many of its roots lie in the interval Δ

$$\alpha < x < \beta.$$

We assume that α, β themselves are not roots of $f(x) = 0$, so that $f(\alpha)$ and $f(\beta)$ are non-zero. For the moment we also assume that $f(x)$ has no multiple root, at least in the interval Δ, so that for no value of this interval can $f(x)$ and $f'(x)$ vanish simultaneously.

Suppose that in some way we can construct a sequence of $m + 1$ continuous functions

$$f(x), f_1(x), f_2(x), \ldots, f_m(x), \tag{1}$$

with the following properties:

1. In the interval Δ, no two consecutive functions f_ν vanish simultaneously.

2. The last of them, $f_m(x)$, does not vanish anywhere in Δ, and hence keeps an unchanging sign.

3. If an intermediate term, say $f_\nu(x)$, vanishes for some x in Δ, then for this x the two adjacent functions $f_{\nu-1}(x)$ and $f_{\nu+1}(x)$ have opposite signs.

4. If $f(x)$ vanishes in the interval Δ, then $f_1(x)$ has, for this value of x, the same sign as $f'(x)$.

We shall call such a sequence (1) a Sturm chain. We shall see later how it is constructed; first we draw some consequences from the definition.

For every value of x for which none of the functions in (1) vanishes, each of these functions has a definite sign. We count one variation of sign whenever, in passing along the chain from left to right, a positive term is followed by a negative one or a negative term by a positive one. Let $V(x)$ denote the number of such variations for a specified value of x. If an intermediate term $f_\nu(x)$ vanishes, then by 3. in passing from $f_{\nu-1}(x)$ to $f_{\nu+1}(x)$ we must count one variation.

Let α, β be two values for which none of the functions f_ν vanishes, and let x increase continuously from α to β. A change in the number of variations can occur only when one of the functions in (1) changes its sign, that is, passes through the value zero, by the assumed continuity.

If this is an intermediate function, then the number of variations is not changed, by 3. Indeed, in

$$f_{\nu-1}, \quad f_\nu, \quad f_{\nu+1}$$

there is always one variation before and after f_ν passes through zero, since $f_{\nu+1}$ and $f_{\nu-1}$ have opposite signs.

But if $f(x)$ passes through zero, then, by 4., one variation is lost. If $f(x)$ passes from negative to positive values, then (compare § 32) $f'(x)$ and hence $f_1(x)$ are positive, so the signs change as follows:

$$\begin{array}{cc} f(x) & f_1(x) \\ - & + \\ + & + \end{array}$$

If $f(x)$ passes from positive to negative values, then $f'(x)$ and $f_1(x)$ are negative, so the signs are

$$\begin{array}{cc} f(x) & f_1(x) \\ + & - \\ - & - \end{array}$$

and in both cases a variation has been lost.

Since $f_m(x)$, by 2., does not change sign, it follows that $V(\alpha) - V(\beta)$ is equal to the number of roots of $f(x) = 0$ lying between α and β, or:

The number of roots of $f(x) = 0$ between α and β is equal to the excess of the number of variations of the chain for $x = \alpha$ over the number of variations for $x = \beta$.

If for $x = \alpha$ or $x = \beta$ one or more of the intermediate functions in (1) vanish, then, by 3., it is immaterial whether we replace the vanishing value by a positive or by a negative one.

§ 87. First example: spherical functions

Before passing to the general methods by which Sturm chains are formed, we shall consider a few examples in which such chains arise from special circumstances.

We first consider the so-called spherical functions, which are frequently used in mathematical physics and mechanics.

By spherical functions one understands a system of integral rational functions of increasing degree defined as follows:

$$\begin{aligned}
&P_0(x) = 1,\\
&P_1(x) = x,\\
&P_2(x) = \frac{3}{2}\left(x^2 - \frac{1}{3}\right),\\
&P_3(x) = \frac{5}{2}\left(x^3 - \frac{3}{5}x\right),\\
&\qquad \cdots\\
&P_n(x) = \frac{1\cdot 3\cdots(2n-1)}{1\cdot 2\cdots n}\left(x^n - \frac{n(n-1)}{2(2n-1)}x^{n-2} + \frac{n(n-1)(n-2)(n-3)}{2\cdot 4(2n-1)(2n-3)}x^{n-4} - \cdots\right).
\end{aligned}$$

Thus $P_n(x)$ is an integral rational function of degree n, containing only even or only odd powers according as n is even or odd. By use of the symbol $\Pi(n)$, the general expression for the spherical functions may also be written

$$P_n(x) = \sum_\mu \frac{(-1)^\mu}{2^n}\frac{\Pi(2n-2\mu)x^{n-2\mu}}{\Pi(\mu)\Pi(n-2\mu)\Pi(n-\mu)}, \tag{1}$$

where the summation over μ is to be extended from $\mu = 0$ as far as $n - 2\mu$ remains non-negative.

Between these functions there hold the following relations, which, after substituting expression (1), are verified by simply comparing the coefficients of equal powers of x:

$$\begin{aligned}
nP_n(x) - (2n-1)xP_{n-1}(x) + (n-1)P_{n-2}(x) &= 0,\\
P_1(x) - xP_0(x) &= 0,
\end{aligned} \tag{2}$$

$$(1-x^2)P_n'(x) + nxP_n(x) - nP_{n-1}(x) = 0. \tag{3}$$

From the last equation it follows, for $x = \pm 1$, that

$$\pm P_n(\pm 1) = P_{n-1}(\pm 1),$$

and hence

$$P_n(1) = 1, \qquad P_n(-1) = (-1)^n. \tag{4}$$

From (2) and (3) it follows that the functions P_n, taken downward from any chosen m,

$$P_m, \quad P_{m-1}, \quad P_{m-2}, \ldots, P_0 \tag{5}$$

have the properties of a Sturm chain in the interval from -1 to $+1$. For if P_n and P_{n-1} vanished simultaneously for some x, then by (2) also P_{n-2}, hence P_{n-3}, and so on down to P_0 would vanish, which is impossible since $P_0 = 1$. Thus condition § 86, 2. is also satisfied for any interval.

If $P_{n-1} = 0$, then by (2) P_n and P_{n-2} have opposite signs, as condition § 86, 3. requires. Equation (3) shows that $P_n(x)$ and $P_n'(x)$ can never vanish simultaneously, since otherwise $P_{n-1}(x)$ would also vanish; and finally (3) also shows that if $P_n(x)$ vanishes and $-1 < x < 1$, then $P_n'(x)$ and $P_{n-1}(x)$ have the same sign.

Now put $x = -1, +1$ in (5). By (4), at $x = -1$ all signs in (5) are alternating, while at $x = +1$ there is no variation. Hence:

The equation $P_m(x) = 0$ has m real roots between $x = -1$ and $x = +1$, and since P_m is of degree m, these are all its roots.

We can draw a somewhat stronger conclusion. Let $\Delta = (\alpha, \beta)$ be an interval containing two, and not more than two, roots ξ, η of $P_m = 0$. Then, in passing from α to β, two variations are lost in the chain (5). We choose α so close to ξ and β so close to η that $P_{m-1}(\alpha)$ has the same sign as

$P_{m-1}(\xi)$, and $P_{m-1}(\beta)$ the same sign as $P_{m-1}(\eta)$. If P_m passes through ξ from negative to positive, then it passes through η from positive to negative; therefore $P'_m(\xi)$, and consequently $P_{m-1}(\xi)$ and $P_{m-1}(\alpha)$, are positive, while $P'_m(\eta)$, and consequently $P_{m-1}(\eta)$ and $P_{m-1}(\beta)$, are negative. The opposite happens if P_m passes through ξ from positive to negative. Thus in passing from α to β in

$$P_m, \quad P_{m-1}$$

one variation is lost. Hence, in the same passage, and consequently also in the passage from ξ to η, in the chain

$$P_{m-1}, \quad P_{m-2}, \dots, P_0$$

one variation must likewise be lost. We conclude:

Between two successive roots of $P_m = 0$ there lies one and only one root of $P_{m-1} = 0$.

§ 88. Second example

As a second example we consider an equation that, for brevity, we shall call the secular equation, because the investigation of the secular perturbations of the planets first led to it.[1] It is also known elsewhere under this name, but it occurs in many other investigations as well, for example in the determination of the principal axes of a surface of second degree and in the theory of small oscillations. Its general analytic significance lies in the fact that it furnishes a special, the so-called orthogonal, transformation of a quadratic form into a sum of squares. We shall take the equation here as it stands, without relation to any particular application, and discuss it by means of our general theorems.

Let $a_{i,k}$, as i and k run through the indices $1, 2, 3, \dots, n$, be any system of real quantities, and suppose that

$$a_{i,k} = a_{k,i}. \tag{1}$$

We consider the symmetric determinant

$$L_n(x) = \begin{vmatrix} a_{1,1} - x & a_{1,2} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} - x & \cdots & a_{2,n} \\ \cdots & \cdots & & \cdots \\ a_{n,1} & a_{n,2} & \cdots & a_{n,n} - x \end{vmatrix}, \tag{2}$$

which is an integral rational function of degree n in x, with coefficient $(-1)^n$ at x^n. The equation $L_n(x) = 0$ is the object of our consideration.

We form the chain of principal minors taken with alternating signs,

$$L_n(x), \quad -L_{n-1}(x), \quad L_{n-2}(x), \dots, \quad (-1)^{n-1} L_1(x), \quad (-1)^n, \tag{3}$$

and shall show that, provided no two consecutive quantities in (3) vanish simultaneously, this is a Sturm chain.

This is a simple consequence of the formula that we already used in § 82 for a similar purpose, and which we may write as

$$L_k S_k - T_k^2 = L_{k-1} L_{k+1}, \tag{4}$$

where S_k and T_k are integral rational functions of x.

We have only to show that the requirements § 86, 1. to 4. are satisfied. Requirement 1. has been included in the assumption, and 2. is satisfied because the last element is equal to ± 1. From (4) it follows that if $L_k = 0$, then L_{k-1} and L_{k+1} have opposite signs, so condition 3. is fulfilled.

[1] Laplace, Histoire de l'Académie des Sciences 1772. Among more recent works one may compare Dziobek, Theorie der Planetenbewegung, Leipzig 1888.

Only condition 4. remains. Applying formula (4) with $k = n-1$, we obtain

$$\frac{\partial L_n}{\partial a_{n,n}} \frac{\partial L_n}{\partial a_{n-1,n-1}} - \left(\frac{\partial L_n}{\partial a_{n,n-1}} \right)^2 = L_n L_{n-2}.$$

It follows that, when $L_n = 0$,

$$\frac{\partial L_n}{\partial a_{n,n}}, \qquad \frac{\partial L_n}{\partial a_{n-1,n-1}}$$

have the same sign; similarly one concludes that all principal minors of L_n,

$$\frac{\partial L_n}{\partial a_{i,i}},$$

have the same sign.

But the derivative of L_n (compare § 22) is

$$L_n'(x) = -\sum_{i=1}^{n} \frac{\partial L_n}{\partial a_{i,i}},$$

and therefore, for a value x at which $L_n(x)$ vanishes, it has the sign opposite to that of L_{n-1}. This is precisely what requirement 4. demands. Hence:

I. If α, β are two real values with $\alpha < \beta$, then the number of roots of $L_n(x)$ lying between α and β is equal to the excess of the number of variations of the chain (3) for $x = \alpha$ over the number of variations for $x = \beta$.

The highest power of x that occurs in $L_k(x)$ is, as already remarked,

$$(-1)^k x^k.$$

If the absolute value of x is sufficiently large, the sign of this term determines the sign of $L_k(x)$. Thus, if for α we take a sufficiently large negative value and for β a sufficiently large positive value, then in (3), for $x = \alpha$, all sign transitions are variations, while for $x = \beta$ all are permanences. Thus there are n roots between α and β, and we obtain:

II. The equation $L_n(x) = 0$ has only real roots.

These two theorems, however, are of only limited applicability as long as we have not freed ourselves from the assumption that in the chain of the L_k no two consecutive terms may vanish simultaneously. This restriction can be removed by the following simple consideration.

Suppose that for some value of x, L_k vanishes but L_{k-1} does not. We can choose the quantities $a_{k+1,i}$, which do not occur in L_k, so that L_{k+1} does not vanish for this value of x; for L_{k+1} cannot vanish identically for all $a_{k+1,i}$, since it contains the term

$$-a_{k+1,k}^2 L_{k-1}$$

[§ 23, (12)].

Therefore, if in the chain

$$L_n, \quad -L_{n-1}, \quad L_{n-2}, \ldots, \quad \pm L_1, \quad \mp 1 \tag{5}$$

several consecutive terms vanish for some value of x, then by changing the $a_{i,k}$ we may derive another sequence

$$L_n', \quad -L_{n-1}', \quad L_{n-2}', \ldots, \quad \pm L_1', \quad \mp 1 \tag{6}$$

in which no two consecutive terms vanish for any value x of the interval $\alpha \ldots \beta$.

At the same time the $a_{i,k}'$, on which the L' depend, can be taken so that they differ from the $a_{i,k}$ by less than any prescribed quantity ω.

If now the numbers α, β are chosen so that no term of the sequence (5) vanishes for $x = \alpha$ or $x = \beta$, then ω can be taken so small that corresponding terms of (5) and (6) have the same signs. By our theorem, the number of roots of $L'_n = 0$ between α and β is determined by the signs of the sequence (6).

On the other hand, ω can also be taken so small that the roots of $L'_n = 0$ differ arbitrarily little from those of $L_n = 0$ (§ 40).

A double root of $L_n = 0$ may, it is true, pass into two simple roots of $L'_n = 0$; but since $L'_n = 0$ has no imaginary roots, these are real. The same applies if $L_n = 0$ has multiple roots.

Thus the theorems I, II remain valid when the assumption that no consecutive terms in the chain of the L_k vanish is dropped; only the multiple roots must then be counted according to their multiplicity.

§ 89. The Sturm functions

After these special examples we turn to the procedure by which a Sturm chain is obtained in all cases.[2] The construction of these functions is in principle extraordinarily simple, although in practical execution it is usually not feasible.

We restrict ourselves here to equations without repeated roots; or we assume that before the application, $f(x)$ has been freed from every common factor with its derivative $f'(x)$.

If we then put

$$f_1(x) = f'(x), \tag{1}$$

then condition § 86, 4. is certainly satisfied. Now we proceed as if we were finding the greatest common divisor of $f(x)$ and $f_1(x)$, but each time reversing the sign of the remainder. Thus by division we form the equations

$$\begin{aligned} f &= q_1 f_1 - f_2, \\ f_1 &= q_2 f_2 - f_3, \\ &\cdots \\ f_{m-2} &= q_{m-1} f_{m-1} - f_m, \end{aligned} \tag{2}$$

where the $q_1, q_2, \ldots, q_{m-1}$, and likewise $f_1, f_2, \ldots, f_m$, are integral rational functions of x. The degrees of the functions $f, f_1, f_2, \ldots, f_m$ decrease, and therefore one can continue the operation until f_m is constant, or at least no longer vanishes in the interval under consideration. That f_m cannot be zero is a consequence of the assumption that f and f_1 have no common divisor.

That in the sequence

$$f, \quad f_1, \quad f_2, \ldots, \quad f_m \tag{3}$$

one actually has a Sturm chain follows immediately by going through the criteria § 86, 1. to 4. For if two consecutive functions of (3) vanished simultaneously, then by (2) all following ones would also vanish; this is impossible because f_m is non-zero.

If $f_\nu(x) = 0$, then from (2) one obtains

$$f_{\nu-1}(x) = -f_{\nu+1}(x),$$

which satisfies all the requirements of § 86.

It is, of course, permissible to multiply the functions in the sequence (3) by positive, for instance constant, factors without ceasing to have a Sturm chain.

[2] Sturm, Mém. sur la résolution des èquations numériques. Mém. de l'Académie de Paris. Sav. ètrang. VI, 1835. Extract in Bull. de Ferussac XI, 1829.

For $n = 2$ we may therefore take the following as the Sturm functions:

$$\begin{aligned} f(x) &= x^2 + ax + b, \\ f_1(x) &= 2x + a, \\ f_2(x) &= a^2 - 4b. \end{aligned}$$